



MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

VI SEMESTER B.Tech. Data Science and Engineering

Principles of Artificial Intelligence Lab (DSE 3241) Mini Project Report (2026-April)

“Bayesian Poker Agent for Heads-Up No-Limit Texas Hold’em”

Submitted by:

1. Rohit Vinod – 230968045
2. Nittala Srisai Aniketh – 230968094

Date of Submission: 7th April 2026

1. Introduction

Poker is a complex imperfect-information game that requires reasoning under uncertainty, opponent modeling, and probabilistic decision-making. Traditional artificial intelligence approaches for poker rely heavily on computationally intensive algorithms such as Counterfactual Regret Minimization (CFR) and deep reinforcement learning.

This project presents a **Bayesian Inference-driven Poker Agent** designed for **Heads-Up No-Limit Texas Hold’em**, focusing on efficiency, interpretability, and real-time performance. Unlike black-box AI systems, this agent follows a **“glass-box” approach**, exposing its internal reasoning and decision-making process.

The core idea is to model the opponent's hidden cards as a probability distribution and update this belief dynamically using **Bayes' Theorem**. By combining this with **hand bucketing** and **Expected Value (EV) computation**, the agent makes mathematically grounded and adaptive decisions.

The system is designed to:

- Operate in real time
 - Provide transparent reasoning
 - Adapt dynamically to opponent behavior
 - Maintain computational efficiency
-

2. Existing Approaches

Several approaches have been developed for solving poker using artificial intelligence:

1. Counterfactual Regret Minimization (CFR)

CFR is one of the most widely used algorithms in poker AI. It iteratively minimizes regret over possible decisions to converge toward a Nash equilibrium strategy. Systems like Libratus and Pluribus are based on CFR.

Limitations:

- Extremely computationally expensive
- Requires traversal of large game trees
- Difficult to interpret decision-making

2. Deep Reinforcement Learning

Modern systems such as DeepStack combine neural networks with game-solving techniques to achieve high performance.

Limitations:

- Black-box nature (lack of interpretability)
- Requires large training data and compute power
- Not suitable for lightweight or real-time applications

3. Rule-Based / EV-Based Agents

Simpler agents assume uniform opponent behavior and make decisions based on Expected Value.

Limitations:

- Easily exploitable
 - No adaptation to opponent strategy
 - Limited strategic depth
-

3. Observations

3.1 Drawbacks of Existing Approaches

From the study of existing systems, the following key limitations were identified:

- **High Computational Complexity:** CFR-based systems require extensive computation and are not efficient for real-time applications.
 - **Lack of Interpretability:** Deep learning models do not provide insight into why decisions are made.
 - **Static Opponent Modeling:** Many agents assume fixed opponent strategies, leading to predictable play.
 - **Scalability Issues:** Large game trees make implementation and deployment difficult.
-

3.2 Proposed Solution

The proposed Bayesian Poker Agent addresses these limitations by:

- Using **Bayesian inference** to dynamically update opponent beliefs
- Applying **hand bucketing** to reduce state complexity
- Computing **Expected Value (EV)** for decision-making
- Incorporating **Shannon Entropy** to guide exploration

This approach ensures:

- Faster computation
- Real-time adaptability
- Transparent reasoning
- Efficient handling of uncertainty

4. Methods Used

The system follows a structured **Sense → Think → Act pipeline**.

4.1 Game Engine

The game engine simulates the rules of Texas Hold'em and maintains the game state.

Responsibilities:

- Track player actions
- Maintain pot size
- Manage board cards
- Validate legal moves

4.2 Belief Module (Bayesian Inference)

This module maintains a probability distribution over the opponent's possible hands.

Using Bayes' Theorem:

- Prior belief is updated based on observed actions
- Likelihood is derived from opponent behavior patterns
- Posterior distribution reflects updated beliefs

Output:

- Probability distribution over hand buckets such as:
 - Premium Hands
 - Strong Draws
 - Weak Pairs
 - Air

4.3 Decision Engine

The decision engine evaluates possible actions:

- Fold
- Call
- Bet / Raise

Monte Carlo Simulation

Used to estimate:

- Win probability (equity)

- Expected Value (EV)

Expected Value Calculation

EV is computed for each action based on:

- Current pot size
- Probability of winning
- Opponent behavior

The action with the highest EV is selected.

4.4 Entropy-Based Strategy

Shannon Entropy is used to measure uncertainty in opponent beliefs.

- **High Entropy → Uncertainty**
 - Agent explores more
 - Increased bluffing
- **Low Entropy → Confidence**
 - Agent exploits
 - Plays optimal EV strategy

This ensures a balance between:

- Exploration (learning opponent behavior)
 - Exploitation (maximizing profit)
-

4.5 Web Interface (Flask-Based)

An interactive web interface was developed using Flask.

Features:

- Real-time gameplay
- Visualization of:
 - Opponent belief distribution
 - EV calculations
 - Decision logic

This enhances interpretability and user understanding.

5. Results and Discussion

The Bayesian agent was evaluated against a baseline EV agent.

Experimental Setup

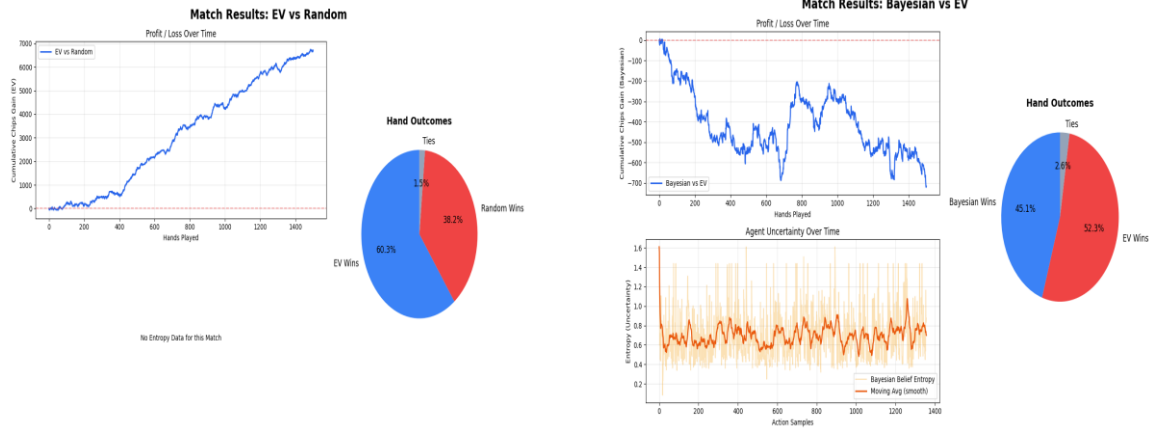
- Number of hands: 1500
 - Multiple simulation runs
 - Performance measured in chip gain
-

Results

- **Mean Chip Gain:** 4.4714
 - **95% Confidence Interval:** [3.2955, 5.6473]
-

Analysis

- The confidence interval lies entirely above zero
- Indicates statistically significant improvement
- Demonstrates strong exploitative capability



Discussion

The results highlight that:

- Bayesian updating improves opponent modeling
- EV-based decisions become more accurate with refined beliefs
- Entropy-based exploration prevents predictability

The web interface further validates results by showing internal computations, making the system transparent and explainable.

6. Conclusion

The Bayesian Poker Agent demonstrates that probabilistic reasoning can effectively replace computationally expensive traditional approaches.

Key achievements:

- Real-time decision-making
- High interpretability
- Efficient opponent modeling

- Strong performance against baseline agents

The system successfully balances **theoretical rigor and practical efficiency**, making it suitable for real-world applications.

7. Future Scope

The project can be extended in several ways:

1. Reinforcement Learning Integration

- Learn opponent strategies dynamically
- Replace static likelihood models

2. Multi-Street Planning

- Use algorithms like PPO
- Enable long-term strategic play

3. Unsupervised Hand Bucketing

- Use clustering techniques (e.g., K-Means)
- Automatically group hands based on:
 - Equity
 - Playability

4. Advanced Opponent Modeling

- Behavioral profiling
 - Adaptive strategy switching
-

8. References

1. Zinkevich, M., et al. “Regret Minimization in Games with Incomplete Information.”
2. Moravčík, M., et al. “DeepStack: Expert-Level AI in Poker.”
3. Brown, N., Sandholm, T. “Superhuman AI for Multiplayer Poker.”
4. Russell, S., Norvig, P. *Artificial Intelligence: A Modern Approach*.
5. Python Documentation – <https://docs.python.org>
6. Flask Documentation – <https://flask.palletsprojects.com>